Design and Implementation of a Modular Dual-Evaporator Transcritical High Temperature Heat Pump

Catalogue of Possible Cycle Configurations

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Date: October 6, 2025

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Abstract

This document presents the various possible cycle configurations for the modular high-temperature heat pump (HTHP) system, along with their corresponding schematics and specified boundary conditions.

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Note on AI assistance

This document was proofread with the assistance of `ChatGPT`, an AI language model developed by `OpenAI`.

1 Subcritical Cycles

1.1 One Evaporator - SC1

The schematic for the One Evaporator Subcritical Heat Pump cycle (SC1) is shown in Figure 1 and the operating conditions are summarized in Table 1.

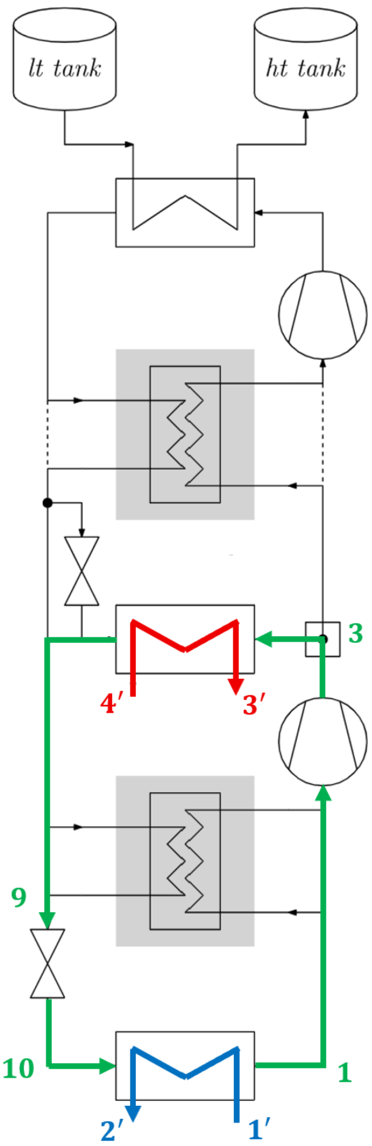

Figure 1: Schematic of the One Evaporator Subcritical Heat Pump cycle (SC1) (*green: working fluid; blue: heat source(s); red: heat sink*)

	Heat source	**Working fluid**	**Heat sink**
Fluid	Water	R290 (propane)	Water
Inlet temperature	$T'_1 = 10\,°\mathrm{C}$	–	$T'_4 = 35\,°\mathrm{C}$
Mass flow rate	$\dot{m}_\mathrm{c} = 0.4\,\mathrm{kg/s}$	–	$\dot{m}_\mathrm{h} = 0.5\,\mathrm{kg/s}$

Table 1: Operating conditions for the SC1 cycle

1.2 One Evaporator with Recuperator - SC1R

The schematic for the One Evaporator Subcritical Heat Pump cycle with Recuperator (SC1R) is shown in Figure 2 and the corresponding operating conditions are summarized in Table 2.

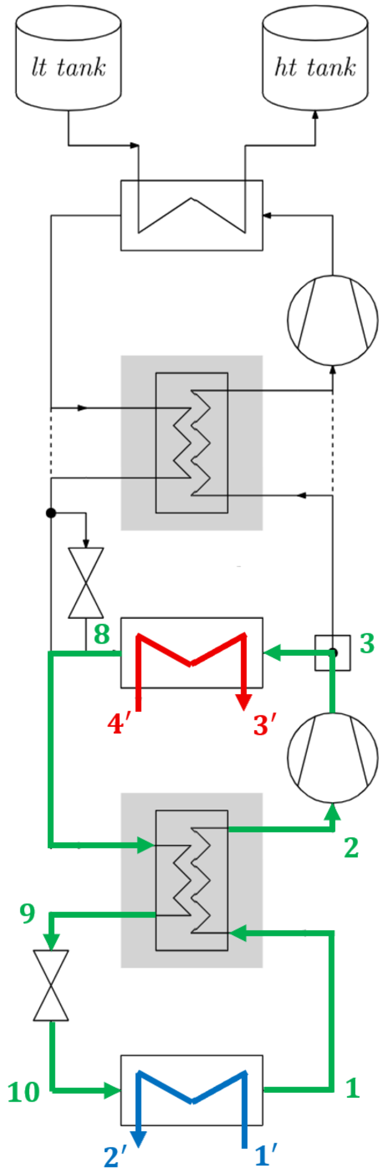

Figure 2: Schematic of the One Evaporator Subcritical Heat Pump cycle with Recuperator (SC1R) (*green: working fluid; blue: heat source(s); red: heat sink*)

	Heat source	**Working fluid**	**Heat sink**
Fluid	Water	R290 (propane)	Water
Inlet temperature	$T_1' = 10\,°\mathrm{C}$	–	$T_4' = 35\,°\mathrm{C}$
Mass flow rate	$\dot{m}_\mathrm{c} = 0.4\,\mathrm{kg/s}$	–	$\dot{m}_\mathrm{h} = 0.5\,\mathrm{kg/s}$

Table 2: Operating conditions for the SC1R cycle

1.3 Two Evaporators - SC2

The schematic for the Dual-Evaporator Subcritical Heat Pump cycle (SC2) is shown in Figure 3 and the corresponding operating conditions are summarized in Table 3.

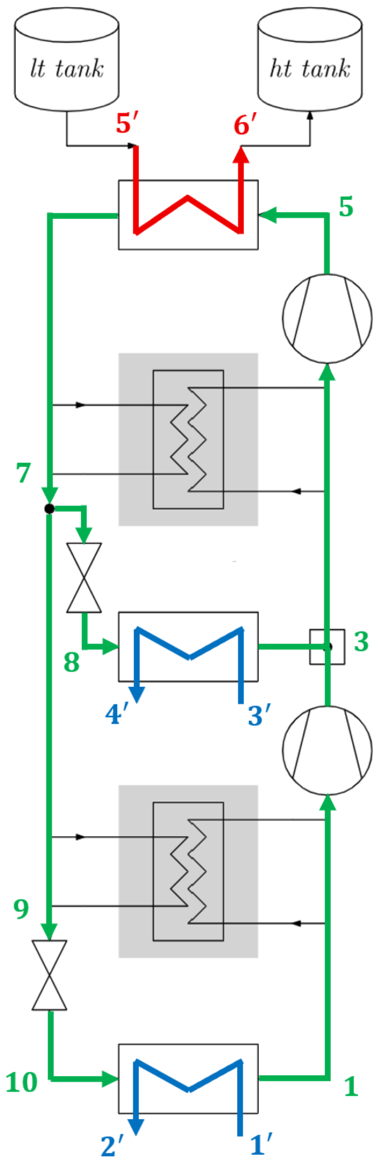

Figure 3: Schematic of the Dual-Evaporator Subcritical Heat Pump cycle (SC2) (*green: working fluid; blue: heat source(s); red: heat sink*)

LT heat source	**MT heat source**	**Working fluid**	**Heat sink**
Water	Water	R290 (propane)	Water
$T_1' = 10\,°\mathrm{C}$	$T_3' = 40\,°\mathrm{C}$	–	$T_5' = 60\,°\mathrm{C}$
$\dot{m}_\mathrm{c} = 0.4\,\mathrm{kg/s}$	$\dot{m}_\mathrm{c} = 1.2\,\mathrm{kg/s}$	–	$\dot{m}_\mathrm{h} = 1\,\mathrm{kg/s}$

Table 3: Operating conditions for the SC2 cycle

1.4 Two Evaporators with Recuperators - SC2R

The schematic for the Dual-Evaporator Subcritical Heat Pump cycle with Recuperators (SC2R) is shown in Figure 4 and the corresponding operating conditions are summarized in Table 4.

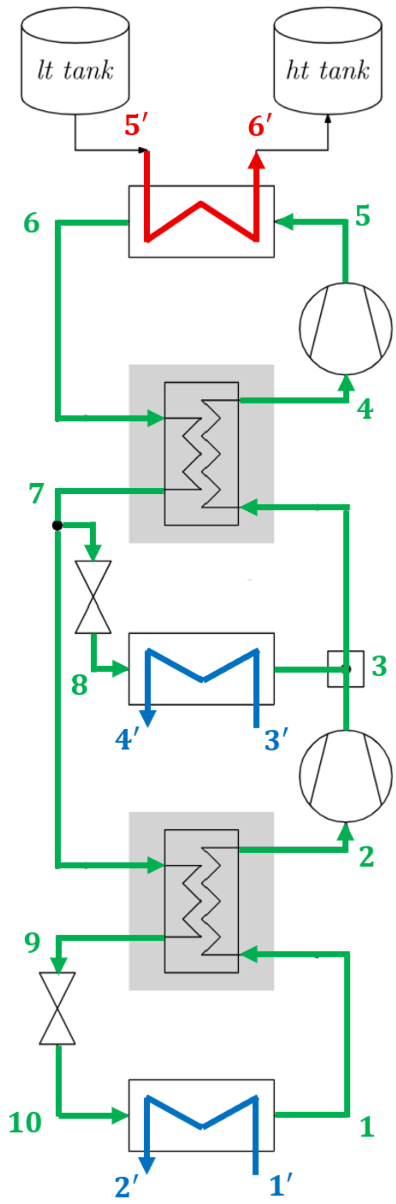

Figure 4: Schematic of the Dual-Evaporator Subcritical Heat Pump cycle with Recuperator (SC2R) (*green: working fluid; blue: heat source(s); red: heat sink*)

LT heat source	MT heat source	Working fluid	Heat sink
Water	Water	R290 (propane)	Water
$T_1' = 10\,°\mathrm{C}$	$T_3' = 40\,°\mathrm{C}$	–	$T_5' = 60\,°\mathrm{C}$
$\dot{m}_\mathrm{c} = 0.4\,\mathrm{kg/s}$	$\dot{m}_\mathrm{c} = 1.2\,\mathrm{kg/s}$	–	$\dot{m}_\mathrm{h} = 1\,\mathrm{kg/s}$

Table 4: Operating conditions for the SC2R cycle

2 Transcritical Cycles

2.1 One Evaporator - TC1

The schematic for the One Evaporator Transcritical Heat Pump cycle (TC1) is shown in Figure 5 and the corresponding operating conditions are summarized in Table 5.

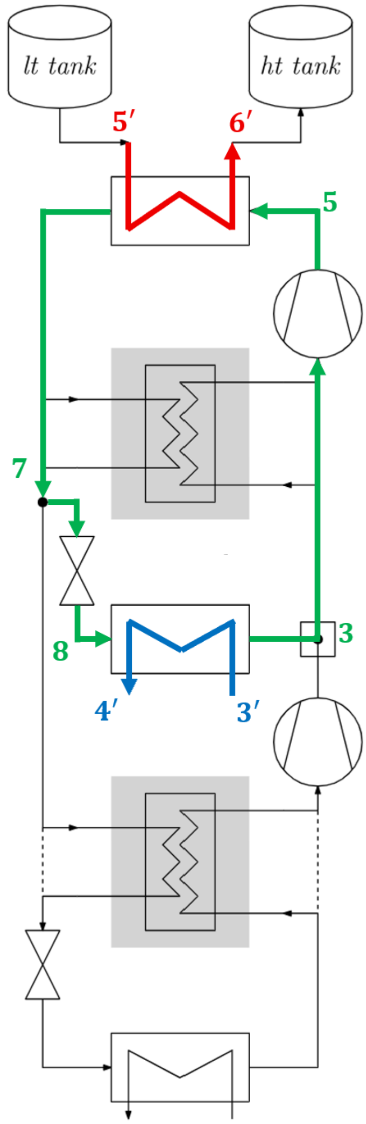

Figure 5: Schematic of the One Evaporator Transcritical Heat Pump cycle (TC1) (*green: working fluid; blue: heat source(s); red: heat sink*)

	Heat source	**Working fluid**	**Heat sink**
Fluid	Water	R290 (propane)	Water
Inlet temperature	$T'_3 = 40\,°C$	–	$T'_5 = 60\,°C$
Mass flow rate	$\dot{m}_c = 0.5\,kg/s$	–	$\dot{m}_h = 0.1\,kg/s$

Table 5: Operating conditions for the TC1 cycle

2.2 One Evaporator with Recuperator - TC1R

The schematic for the One Evaporator Transcritical Heat Pump cycle with Recuperator (TC1R) is shown in Figure 6 and the corresponding operating conditions are summarized in Table 6.

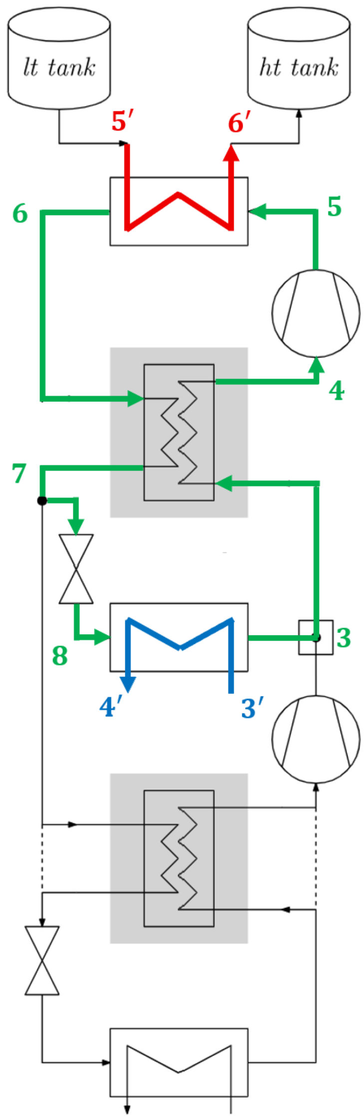

Figure 6: Schematic of the One Evaporator Transcritical Heat Pump cycle with Recuperator (TC1R) (*green: working fluid; blue: heat source(s); red: heat sink*)

	Heat source	**Working fluid**	**Heat sink**
Fluid	Water	R290 (propane)	Water
Inlet temperature	$T_3' = 40\,°\text{C}$	–	$T_5' = 60\,°\text{C}$
Mass flow rate	$\dot{m}_\text{c} = 0.5\,\text{kg/s}$	–	$\dot{m}_\text{h} = 0.1\,\text{kg/s}$

Table 6: Operating conditions for the TC1R cycle

2.3 Two Evaporators - TC2

The schematic for the Dual-Evaporator Transcritical Heat Pump cycle (TC2) is shown in Figure 7 and the corresponding operating conditions are summarized in Table 7.

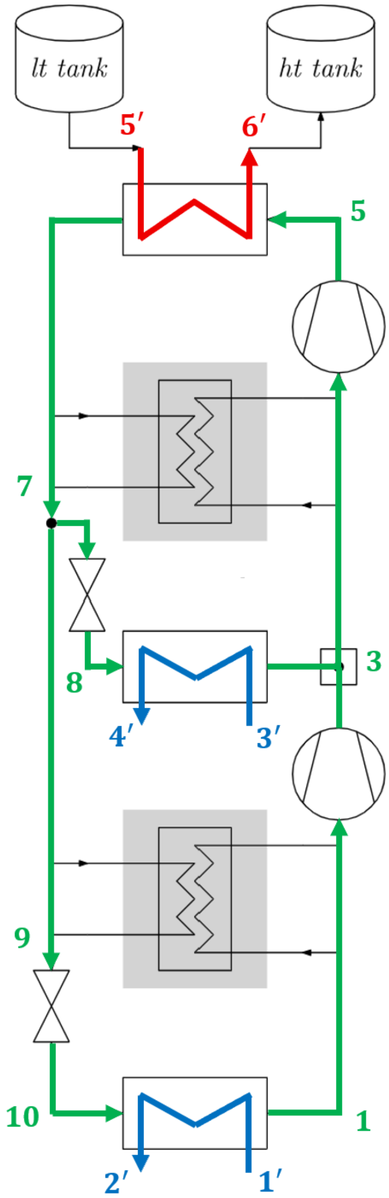

Figure 7: Schematic of the Dual-Evaporator Transcritical Heat Pump cycle (TC2) (*green: working fluid; blue: heat source(s); red: heat sink*)

LT heat source	**MT heat source**	**Working fluid**	**Heat sink**
Water	Water	R290 (propane)	Water
$T_1' = 10\,°\mathrm{C}$	$T_3' = 40\,°\mathrm{C}$	–	$T_5' = 60\,°\mathrm{C}$
$\dot{m}_\mathrm{c} = 0.25\,\mathrm{kg/s}$	$\dot{m}_\mathrm{c} = 0.75\,\mathrm{kg/s}$	–	$\dot{m}_\mathrm{h} = 0.15\,\mathrm{kg/s}$

Table 7: Operating conditions for the TC2 cycle

2.4 Two Evaporators with Recuperators - TC2R

The schematic for the Dual-Evaporator Transcritical Heat Pump cycle with Recuperators (TC2R) is shown in Figure 8 and the corresponding operating conditions are summarized in Table 8.

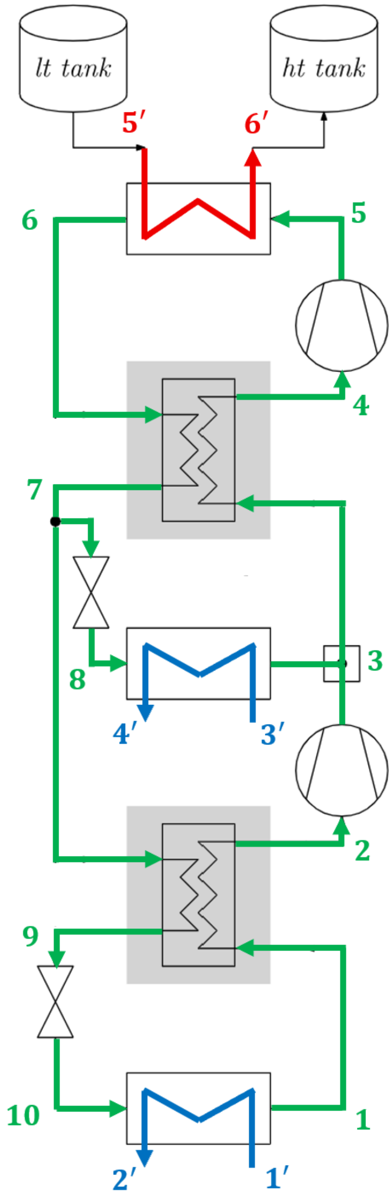

Figure 8: Schematic of the Dual-Evaporator Transcritical Heat Pump cycle with Recuperator (TC2R) (*green: working fluid; blue: heat source(s); red: heat sink*)

LT heat source	MT heat source	Working fluid	Heat sink
Water	Water	R290 (propane)	Water
$T_1' = 10\,°\mathrm{C}$	$T_3' = 40\,°\mathrm{C}$	–	$T_5' = 60\,°\mathrm{C}$
$\dot{m}_\mathrm{c} = 0.25\,\mathrm{kg/s}$	$\dot{m}_\mathrm{c} = 0.75\,\mathrm{kg/s}$	–	$\dot{m}_\mathrm{h} = 0.15\,\mathrm{kg/s}$

Table 8: Operating conditions for the TC2R cycle

3 Notes

One can note the following regarding the operating conditions of the cycles:

- For cycles with two evaporators, the mass flow rate at the low-temperature heat source is always one-third of the mass flow rate at the medium-temperature heat source. Assuming the same temperature difference at both evaporators, this ensures that $\beta \approx 75\%$.
- For transcritical cycles, the mass flow rate at the heat sink is always lower than for subcritical cycles. This adjustment increases the temperature glide at the heat sink, highlighting one of the key advantages of transcritical operation.